

Untitled

Purpose. This document constitutes a complete technical specification for the AI-assisted generation of Quarto Revealjs presentation slides (.qmd) for a university-level introductory statistics course. All formatting conventions, structural requirements, and code standards described herein must be strictly observed. Before producing any slide content, the AI agent is expected to read this document in its entirety.

YAML Front Matter

Every slide file must begin with the following YAML header. Only the `title`, `subtitle` (reflecting the chapter number and name), and the language-switch hyperlink (`chX-zh.html`) are to be modified. All other fields must remain unchanged.

```
---
title: "Statistics"
subtitle: "Chapter X: <Chapter Title>"
author: "Yu-You Liou"
institute: "Shih Chien University"
date: "2026-06-04"
format:
  revealjs:
    html-math-method: katex
    smaller: true
    theme: serif
    footer: "Elementary Statistics: A Step by Step Approach"
    logo: "logo.svg"
    editor: source
    slide-number: true
    scrollable: false
```

```

transition: "none"
menu:
  useTextContentForMissingTitles: false
  hideMissingTitles: true
  side: 'left'
  titleSelector: 'h1'
title-slide-attributes:
  data-notes: ""
include-after-body:
  text: |
    <script>
    Reveal.on('ready', function() {
      var title = Reveal.getSlides()[0];
      if (title) {
        title.insertAdjacentHTML('beforeend',
          '<p style="font-size:0.5em; margin-top:1.5em;">' +
          '<a href="chX-zh.html" style="color:#555;">[  ]</a>' +
          '</p>');
      }
    });
    </script>

```

Note. Replace X in `chX-zh.html` with the appropriate chapter number (e.g., `ch3-zh.html` for Chapter 3).

Immediately following the YAML header, include a global R setup chunk:

```

```{r setup, include=FALSE}
library(tidyverse)
```

```

Overall Slide Structure

The slides within each chapter file must appear in the following mandatory sequence:

Title slide (auto-generated from YAML)

↓

```

## Overview
↓
# Section X-1: <Name> {background-color="#2c4a6e"}
  [Section content: definitions, rules, examples, etc.]
↓
# Section X-2: <Name> {background-color="#2c4a6e"}
  [Section content]
↓
  (Additional sections as required)
↓
## Key Formulas    (include if the chapter contains multiple key formulae)
↓
## Key Takeaways
↓
## Acknowledgement

```

The Overview Slide

The slide immediately following the title slide must be an **Overview** slide. It consists of a brief introductory paragraph describing the chapter's central theme, followed by a two-column Markdown table summarising the content of each section.

```

## Overview

Chapter X introduces <brief description of the chapter's central theme>.

Section	Topics
X-1	<Key topics of Section X-1>
X-2	<Key topics of Section X-2>
X-3	<Key topics of Section X-3>

```

Constraint. The table must contain exactly two columns: **Section** and **Topics**. No additional columns are permitted.

Section Divider Slides

Each new section must begin with a dedicated title-divider slide using a **level-one heading** (#) with a dark-blue background:

```
# Section X-Y: <Section Title> {background-color="#2c4a6e"}
```

Requirements:

- Use a single #, not ##.
- The `background-color` attribute must always be `"#2c4a6e"`.
- The section title follows the colon immediately, e.g., `Section 2-1: Organizing Data`.

General Content Slides

All content slides following a section divider use a **level-two heading** (##):

```
## <Slide Title>
```

```
<Prose, bullet lists, bold/italic emphasis - standard Markdown formatting>
```

When content exceeds a single slide, use a bare horizontal rule (---) to create a new slide. The continuation slide may omit a heading or carry a repeated heading as appropriate.

Definitions

Definitions of statistical terminology must be formatted using the `callout-note` block:

```
:::{.callout-note}
```

```
## <Term Name>
```

```
<Definition text. Key terms are typeset in bold; mathematical expressions use LaTeX notation>
```

```
:::
```

Example:

```
:::{.callout-note}  
## Frequency Distribution  
  
A frequency distribution is a table that organizes data values (or ranges of values)  
along with the count of how often each value or range occurs in the dataset.  
:::
```

Important Rules and Guidelines

Formal rules, construction guidelines, and methodological constraints must be formatted using the `callout-important` block:

```
:::{.callout-important}  
## <Guideline Title>  
  
1. <First rule; key terms in bold.>  
2. <Second rule.>  
3. <Additional rules as needed.>  
  
Formula:  $\text{<LaTeX expression>}$   
:::
```

Example:

```
:::{.callout-important}  
## Six Construction Guidelines  
  
1. Use between 5 and 20 classes - enough to show pattern without excessive detail.  
2. The class width should be an odd number when possible so midpoints are whole numbers.  
3. Classes must be mutually exclusive - each data value belongs to exactly one class.  
4. Classes must be continuous - no gaps between classes, even if some have zero frequency.  
5. Classes must be exhaustive - every data value must fall into some class.  
6. Classes should be equal in width (except possibly the first or last class).  
  
Formula:  $\text{Class width} \approx \frac{\text{highest value} - \text{lowest value}}{\text{number of classes}}$   
:::
```

Worked Examples

Question Slide

- The slide title must include a named anchor using the syntax `{#example-X-Y}` to enable cross-referencing.
- The problem statement is enclosed in a `callout-warning` block.
- The callout heading states the example's descriptive title.
- If this example is referenced by a later example, include a back-navigation link at the bottom of the callout.
- Raw data arranged in a tabular layout must be wrapped in a fenced code block (triple backticks, no language tag).

```
## Example X-Y {#example-X-Y}

:::{.callout-warning}
## <Example Title>

<Problem statement.>
```

Back to [Example X-Z]({#example-X-Z}) ← include only when this example is cited elsewhere
:::

Solution Slide(s)

- Separate the solution from the question using a bare ---.
- Repeat the heading `## Example X-Y` without a new anchor.
- Enclose the solution in a `callout-tip` block with the heading `## Solution`.
- When a solution requires multiple slides (e.g., data preparation, plotting, output), use additional --- separators. On each continuation slide, repeat the `## Example X-Y` heading and open a new `callout-tip` block.
- **Each step within a solution must be preceded by a bold descriptor** indicating what is being accomplished on that slide (e.g., **`**Prepare data**`**, **`**Draw figure**`**, **`**Output figure**`**).

```
---

## Example X-Y

:::{.callout-tip}
```

```
## Solution
```

```
**<Step description>**
```

```
::: {.cell layout-align="center"}
```

```
```{r .cell-code}
```

```
R code
```

```
:::
```

```
:::
```

When the solution continues on a subsequent slide:

```
```markdown
```

```
---
```

```
## Example X-Y
```

```
::: {.callout-tip}
```

```
## Solution
```

```
**<Next step description>**
```

```
::: {.cell layout-align="center"}
```

```
```{r .cell-code}
```

```
R code
```

```
:::
```

```
:::
```

> **R chunk label convention.** Labels follow the pattern ``e<chapter>-<example>-<slide sequence>``. Labels must be globally unique within the file.

```
Caution Blocks
```

When an example involves a boundary condition, approximation caveat, or methodological warning, use a `warning`` block (distinct from the problem statement):

```
```markdown
:::{.callout-warning}
## Caution
<Warning text.>
:::
```

Cross-Example Navigation

When a later example cites data or results from a prior example, the reference in the problem statement must be a hyperlink:

Using the blood type data from [Example X-Y] (#example-X-Y), construct a pie graph.

Correspondingly, the **cited example's question slide** must include a back-navigation link at the bottom of its `callout-warning` block:

Back to [Example X-Z] (#example-X-Z)

This bidirectional linking ensures that readers can navigate freely between related examples.

Mathematical Notation

All mathematical symbols, variables, and expressions must be typeset in LaTeX. The use of Unicode characters as substitutes is strictly prohibited.

Prohibited	Required
$n \geq 30$	$n \geq 30$
μ (Unicode)	μ
σ (Unicode)	σ

Prohibited	Required
× (Unicode)	$\times$
% inside an equation	$\% = \frac{f}{n} \times 100$

Inline expressions use $\dots$; display-mode equations use
$$\dots$$
.

Examples:

The class width is approximated as $\text{Class width} \approx \frac{H - L}{k}$.

$\% = \frac{f}{n} \times 100$

Key Formulas Slide

If a chapter contains multiple formulae of central importance, include a dedicated **Key Formulas** slide immediately before the Key Takeaways slide:

```
## Key Formulas

:::{.callout-important}
## Important Formulas


$$\text{Formula one}$$



$$\text{Formula two}$$



$$\text{Formula three}$$

:::
```

Key Takeaways Slide

This slide is **mandatory** and must appear as the penultimate slide. Each takeaway is a concise declarative statement. Mathematical notation must use LaTeX.

Key Takeaways

1. <First key insight.>
2. <Second key insight.>
3. <Third key insight, e.g., the CLT applies when $n \geq 30$.>

Acknowledgement Slide

This slide is **mandatory** and must appear as the final slide. The following text is fixed and must not be altered:

Acknowledgement

- **Copyright notice.** These teaching materials have been adapted from **Bluman, A. G. (2010)**.
- **Non-commercial use only.** These materials are strictly intended for educational purposes.
- **Proper attribution.** Any reproduction, distribution, or use of these materials must properly acknowledge the source.

Slide Pagination Rules

- Use a bare --- (three hyphens on a line by themselves) to separate slides.
- When continuing the same example across multiple slides, **do not introduce a new anchor** — simply repeat the example heading.
- When introducing entirely new content, open a new ## heading.

R Code Chunk Defaults

Unless a specific slide requires different settings, all R chunks must use the following options:

```
echo = T, warning = FALSE, out.width = "80%", fig.align = 'center'
```

Tabular output must be rendered using:

```
knitr::kable(df, align = "c")
```

Complete Structural Template

The following skeleton illustrates the required structure for a complete chapter file:

```
---
title: "Statistics"
subtitle: "Chapter X: <Chapter Title>"
author: "Yu-You Liou"
... (complete YAML as specified in Section 1)
---

## Overview

Chapter X introduces ...

| Section | Topics |
|-----|-----|
| X-1 | ... |
| X-2 | ... |

# Section X-1: <Section Title> {background-color="#2c4a6e"}

## <Content Slide Title>

<Prose explanation.>
```

```

:::{.callout-note}
## <Term>
<Definition.>
:::

---

## Example X-1 {#example-X-1}

:::{.callout-warning}
## <Example Title>
<Problem statement.>
:::

---

## Example X-1

:::{.callout-tip}
## Solution

**Prepare data**

::: {.cell layout-align="center"}

```{.r .cell-code}
R code
```

:::

:::

---

## Example X-1

:::{.callout-tip}
## Solution

**Output figure**

```

```

::: {.cell layout-align="center"}

```{r .cell-code}
R code
```

:::

:::

# Section X-2: <Section Title> {background-color="#2c4a6e"}

... (additional sections)

## Key Formulas

::: {.callout-important}
## Important Formulas
 $\text{Formula}$ 
:::

## Key Takeaways

1. ...
2. ...

## Acknowledgement

- Copyright notice. These teaching materials have been adapted from Bluman, A. G. (2010)
- Non-commercial use only. These materials are strictly intended for educational purposes
- Proper attribution. Any reproduction, distribution, or use of these materials must properly

```

End of specification. Generate the .qmd slide file in strict accordance with all rules set out above.